



Strength of Materials II

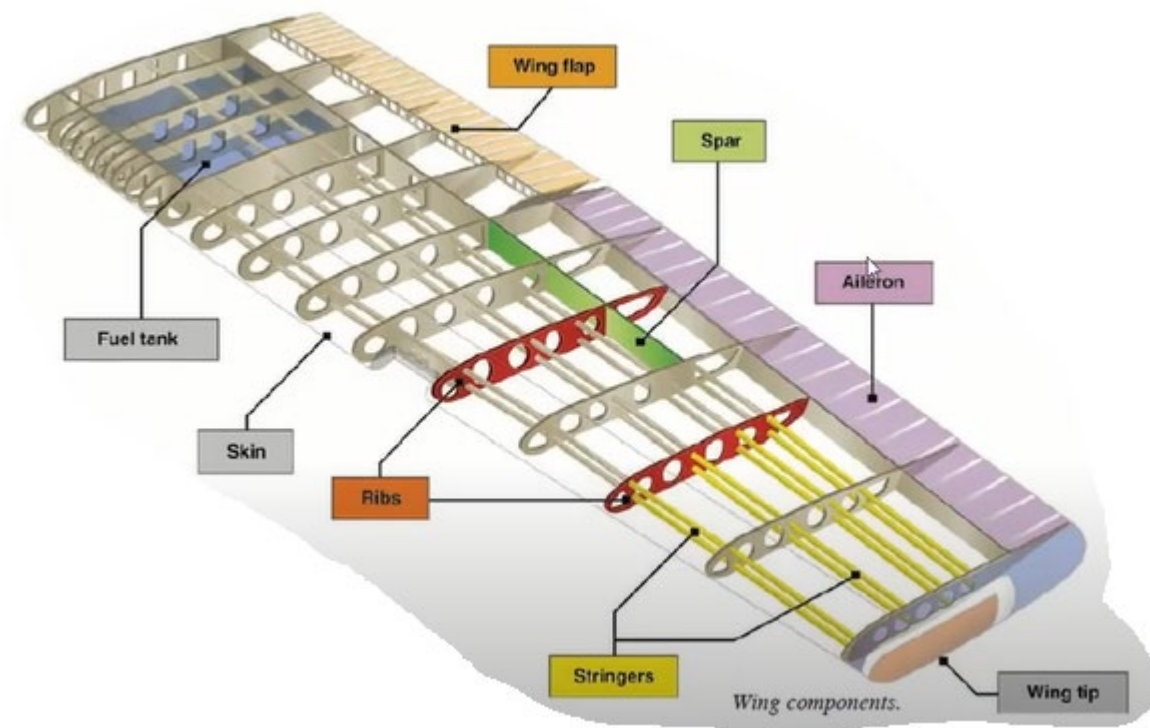
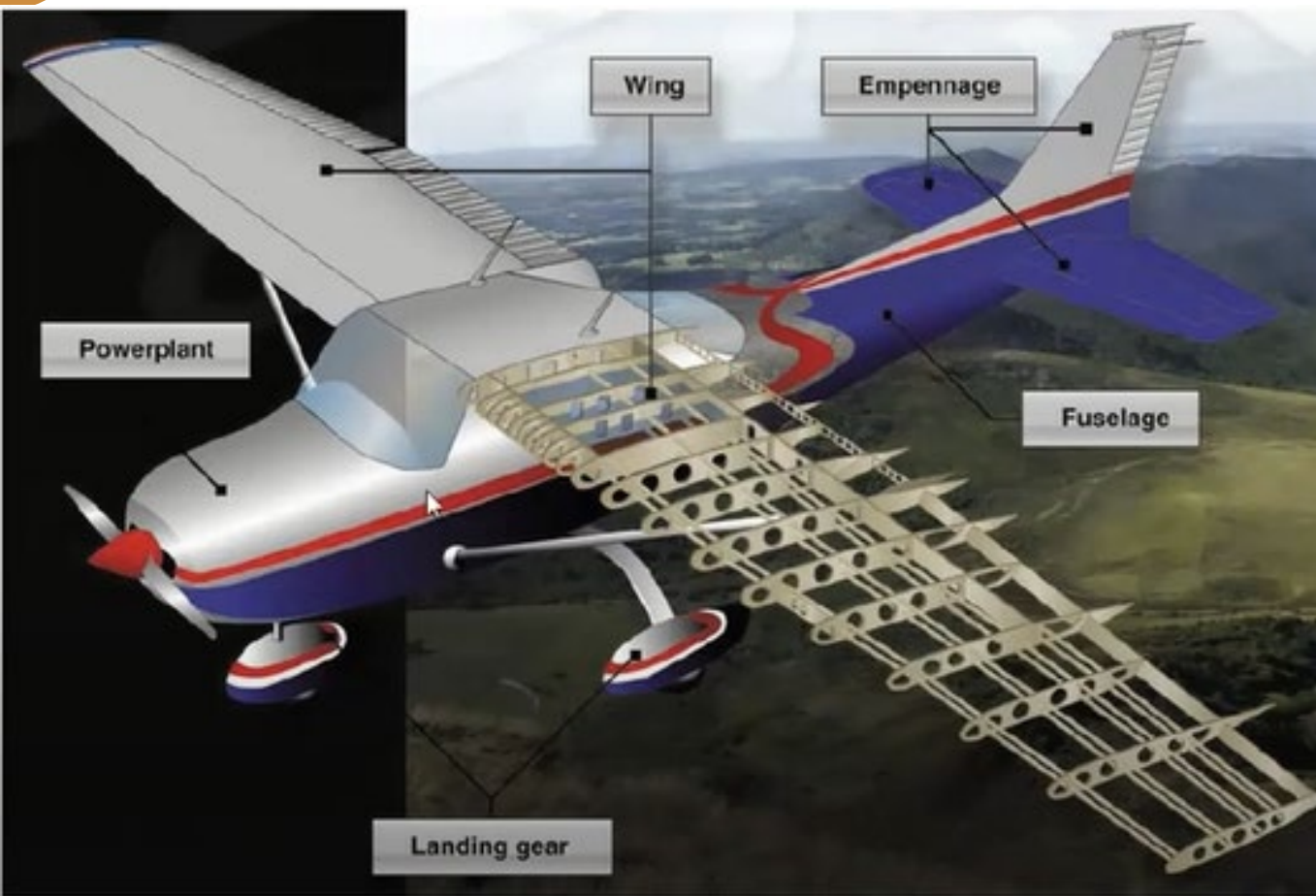


Chapter Six: Unsymmetrical Bending

MEng3111



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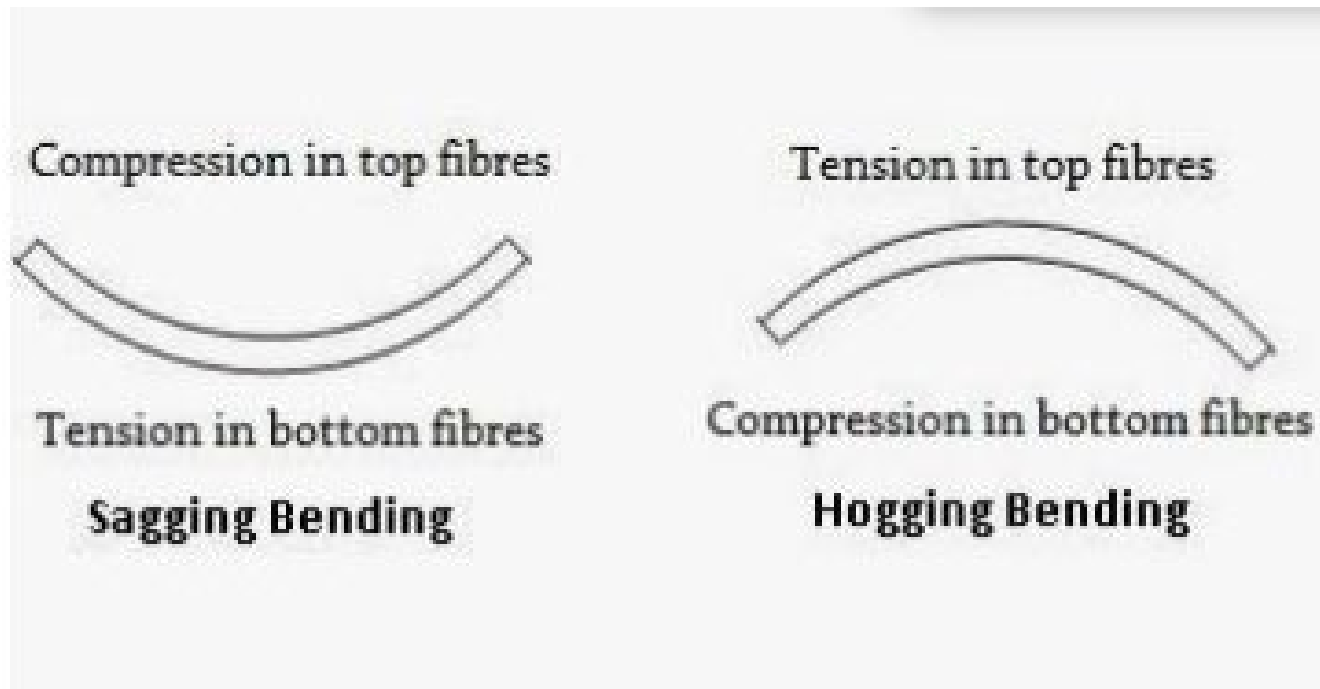
Introduction

- A structural system is a device which can resist, carry and transmit the externally applied loads to the neighboring structural members, without causing any permanent deformation
- A structural component must be capable enough to resist loads due to tension, compression, bending, shear and torsion or any combination.
- Structural members can be of open section beam and closed section beam, both type of beams are subjected to axial, bending, shear and torsional loads.

Bending of Beams

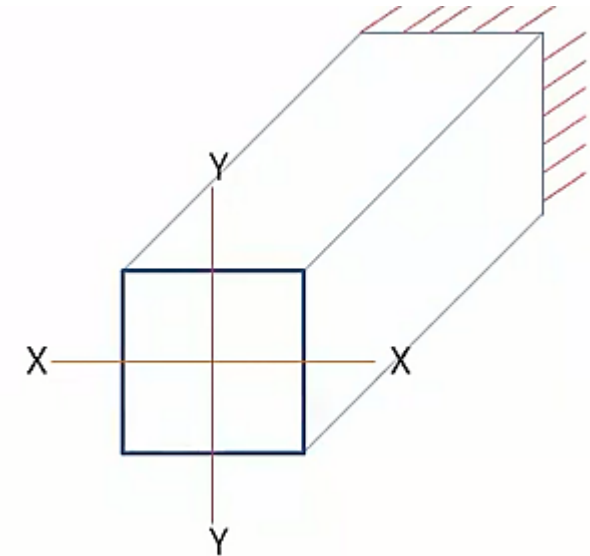
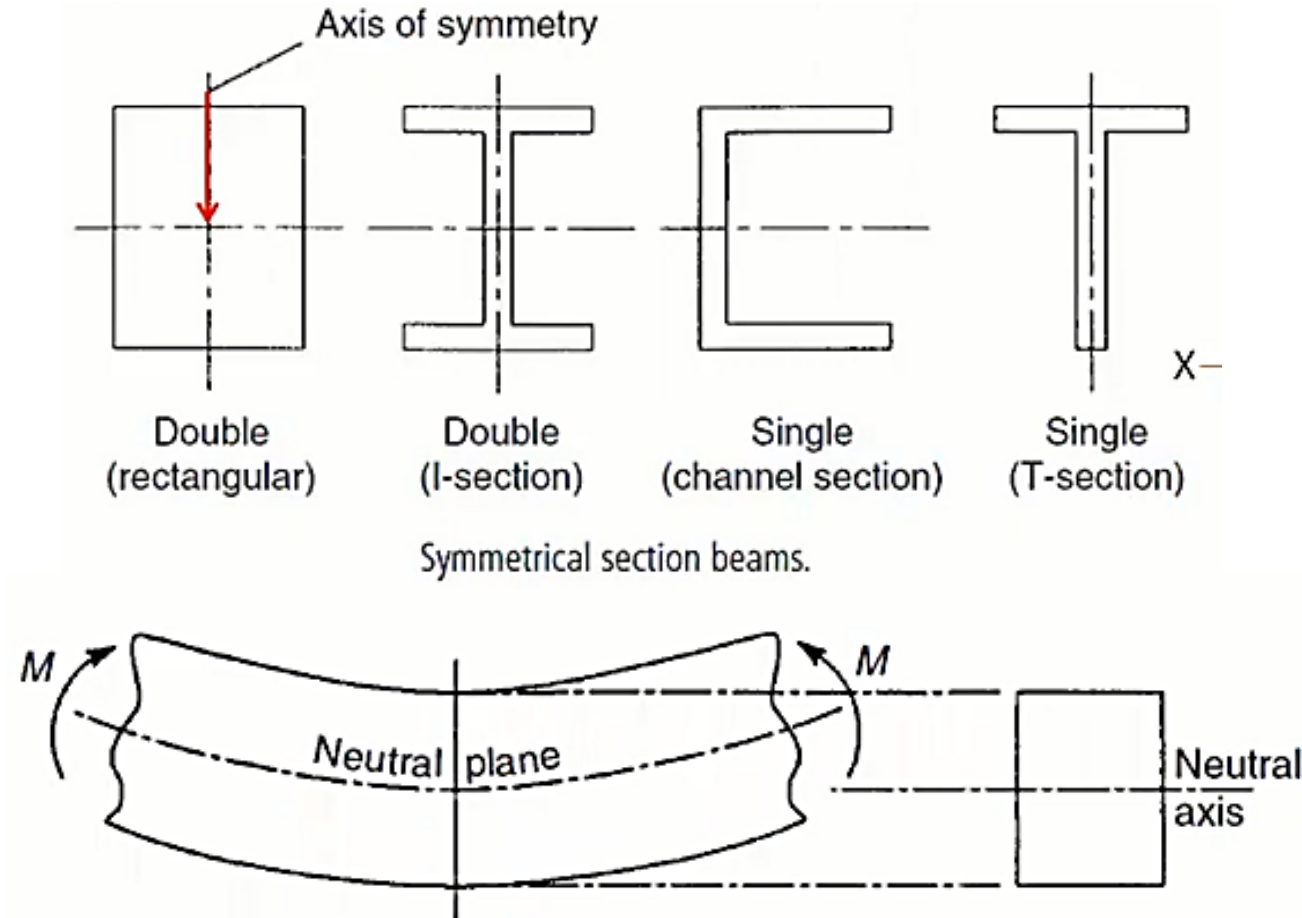
- When an axial loads is applied to a member, it produces a uniform direct stress (load/Area) and an axial displacement.
- A situation is different when applied loads causes a beam to bend which, if the loads are vertical; and causes the beam to sag or hog.

‘()’ ‘()’



Symmetrical bending

- Symmetrical bending arises in beams which have either singly or doubly symmetrical cross-sections.



$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma_z}{y}$$

A) Beam subjected to a pure sagging bending moment

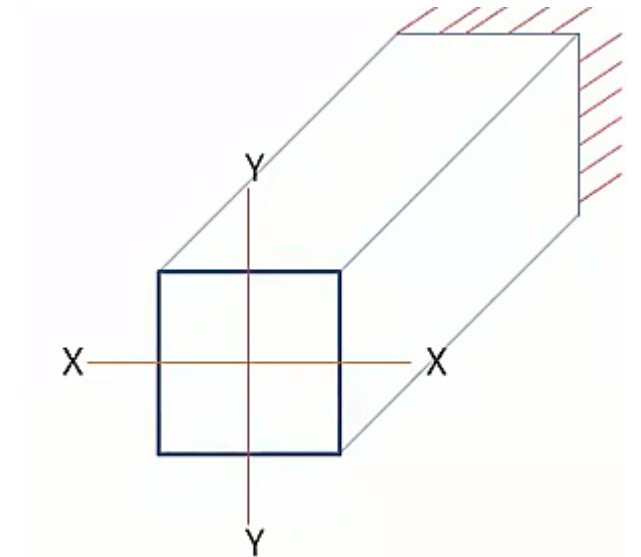
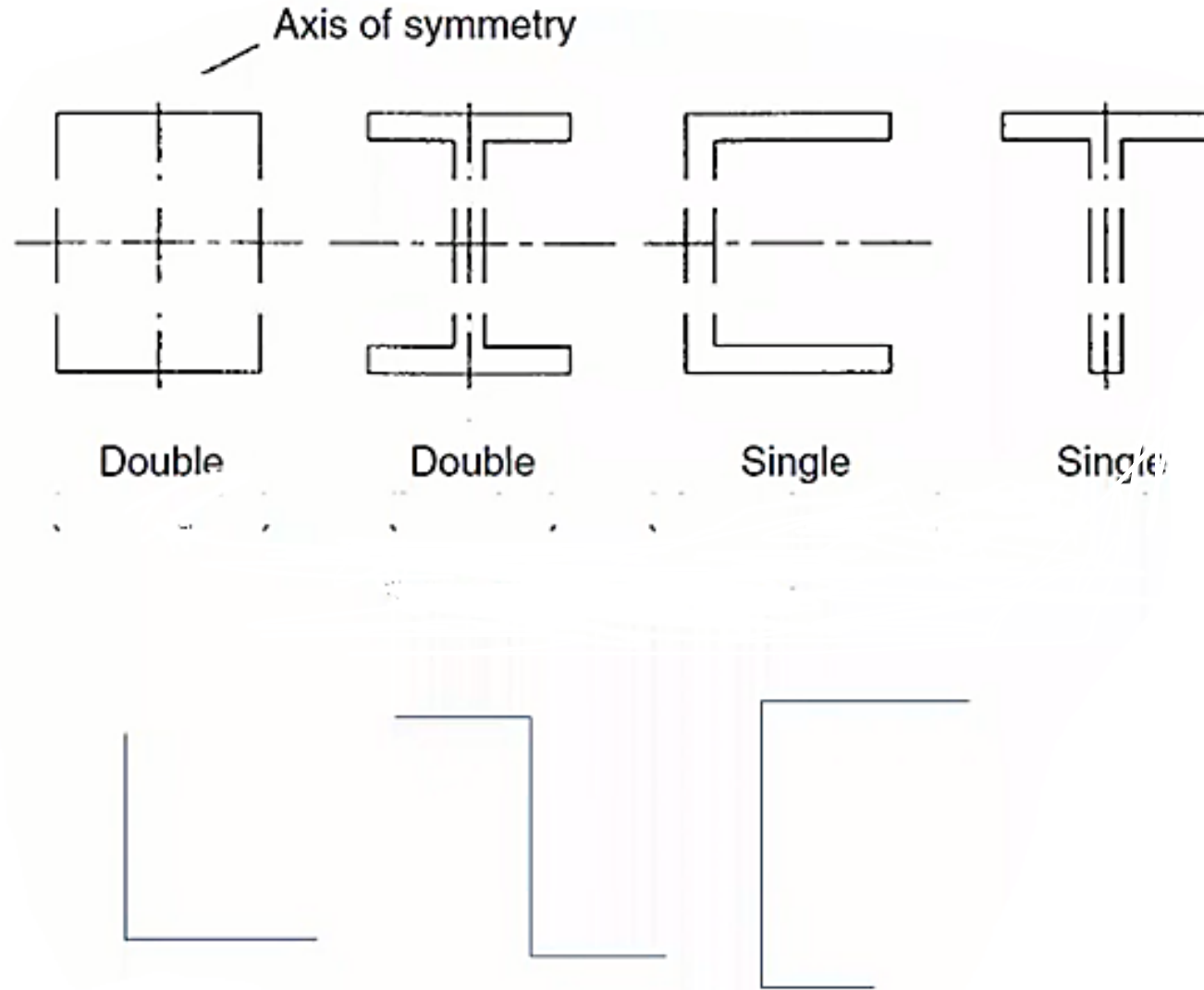
Bending Assumptions

- The material of the beam is linearly elastic, i.e. it obeys Hooke's law, and that the material of the beam is homogeneous.
- In pure bending, the plane remains plane before and after bending.

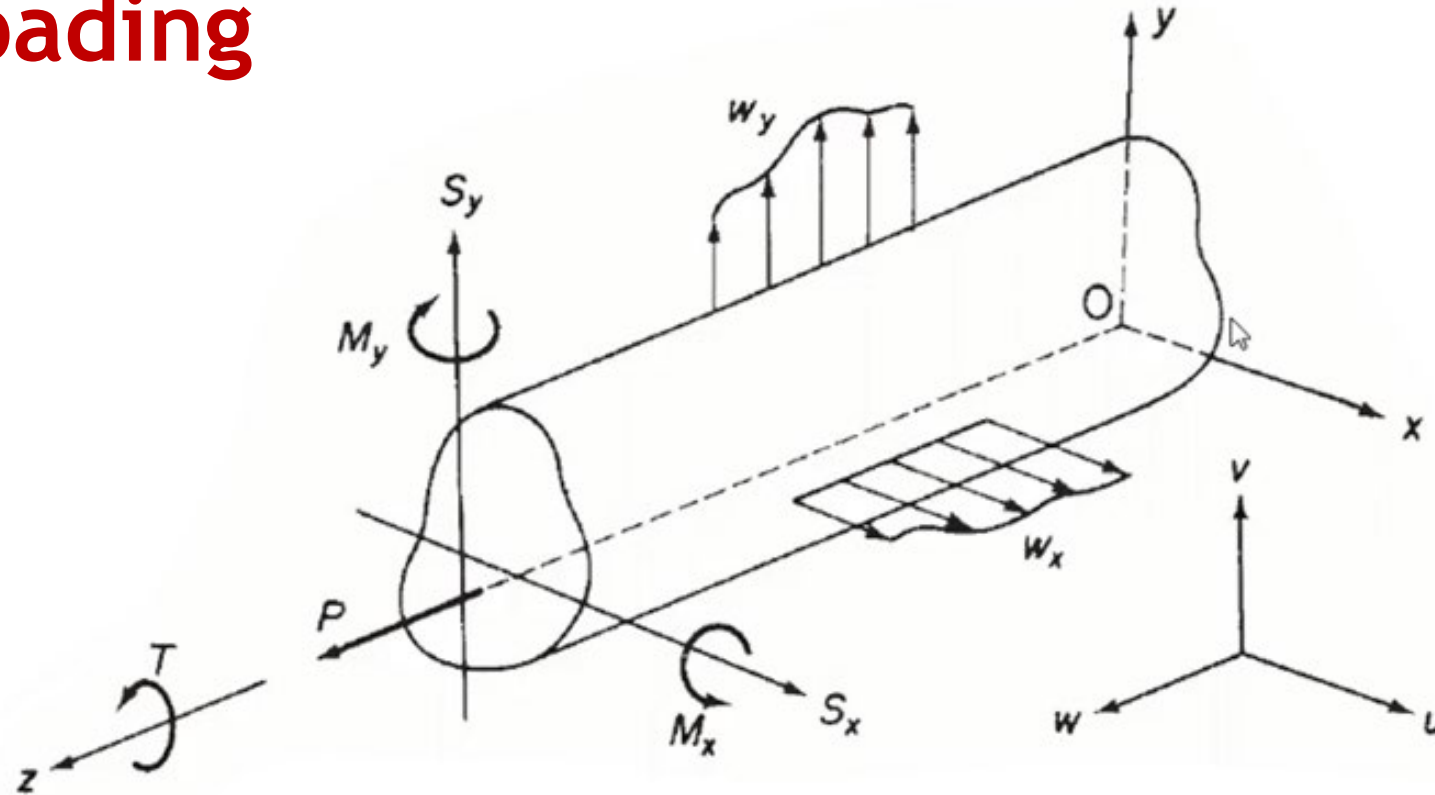
Pure Bending Stress expression: $\frac{M}{I} = \frac{E}{R} = \frac{\sigma_z}{y}$

Unsymmetrical bending

Unsymmetrical bending arises in beams which have either singly or doubly symmetrical cross-sections and subjected to skew loads.

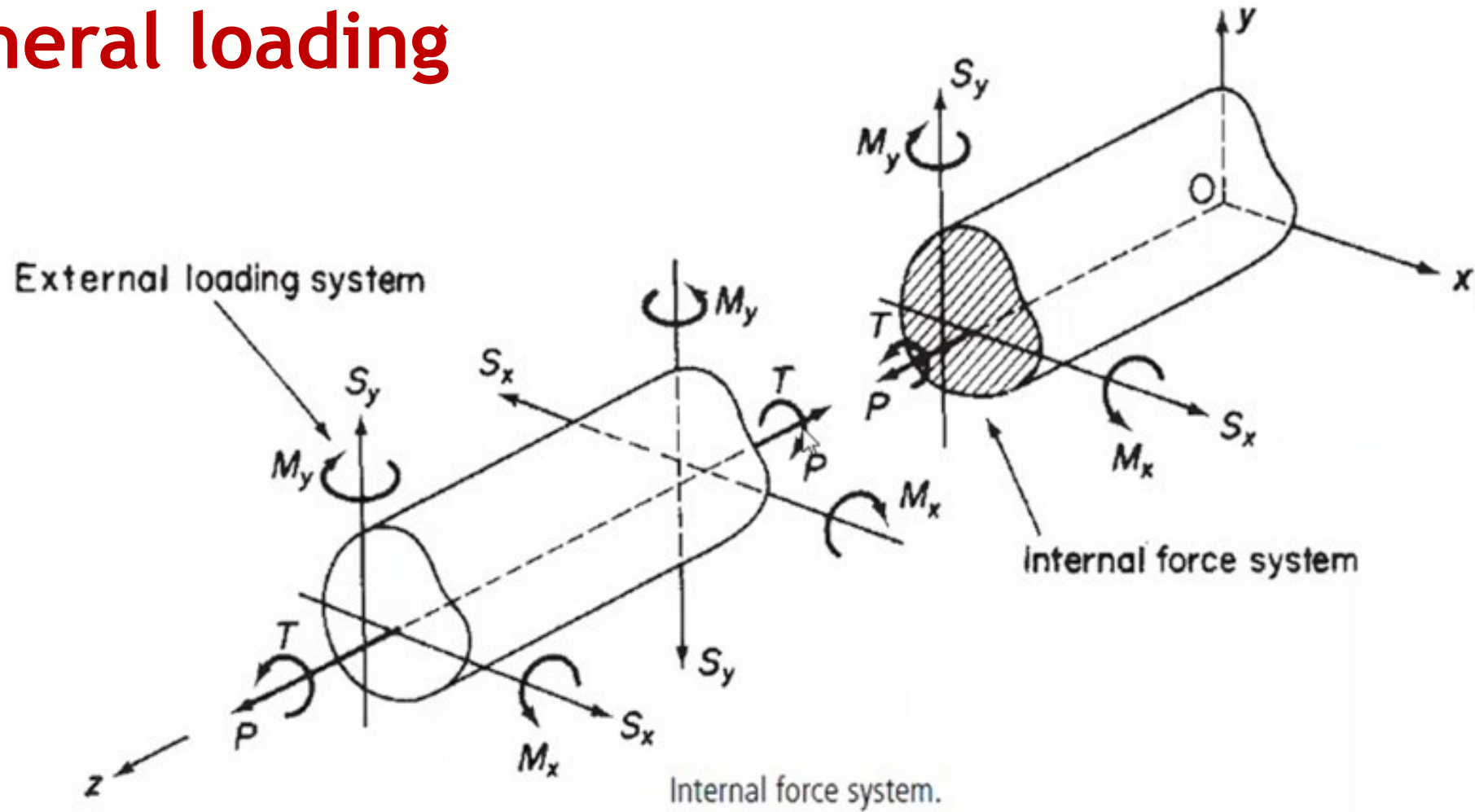


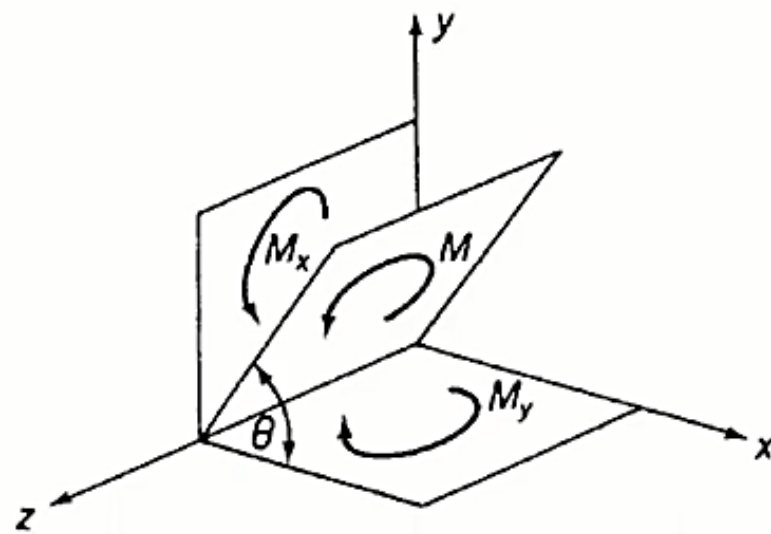
General loading



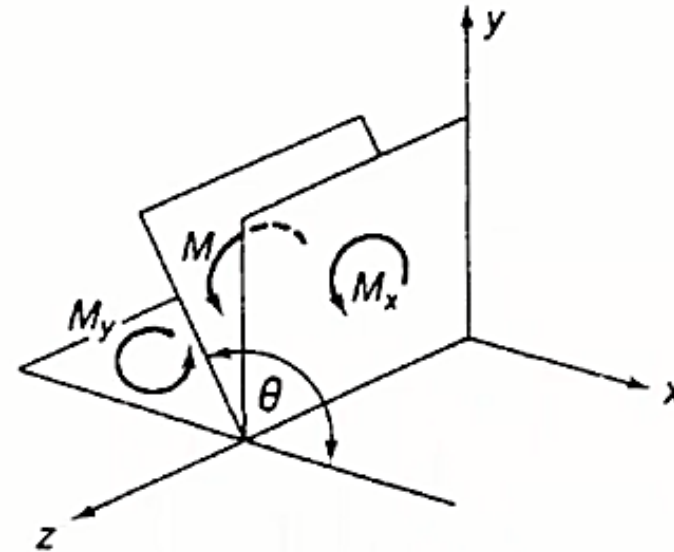
Notation and sign convention for forces, moments and displacements.

General loading





(a) $\theta < 90^\circ$



(b) $\theta > 90^\circ$

Resolution of bending moments.

for $\theta < \pi/2$, M_x and M_y positive

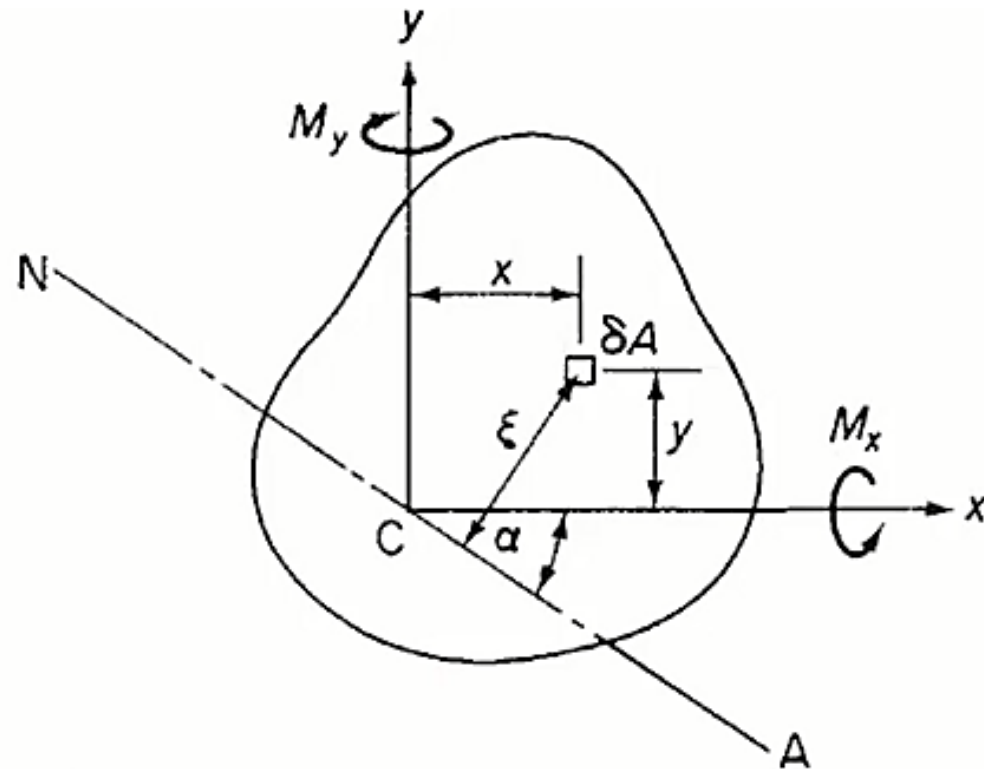
for $\theta > \pi/2$, M_x positive and M_y negative

In both cases, for the sense of M shown

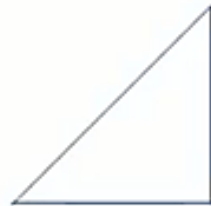
$$M_x = M \sin \theta$$

$$M_y = M \cos \theta$$

Direct stress distribution due to bending



Let ρ be the radius of curvature



From solid mechanics,

$$\sigma = Ee$$

$$\sigma_z = E\varepsilon_z \text{ ----- (1)}$$

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma_z}{y}$$

$$\frac{M}{I} = \frac{y}{R} = \frac{\sigma_z}{E}$$

$$\varepsilon_z = \frac{\xi}{\rho} \text{ ----- (2)}$$

$$\sigma_z = E \frac{\xi}{\rho} \text{ ----- (3)}$$

$$\xi = x \sin \alpha + y \cos \alpha$$

$$\sigma_z = \frac{E}{\rho}(x \sin \alpha + y \cos \alpha)$$

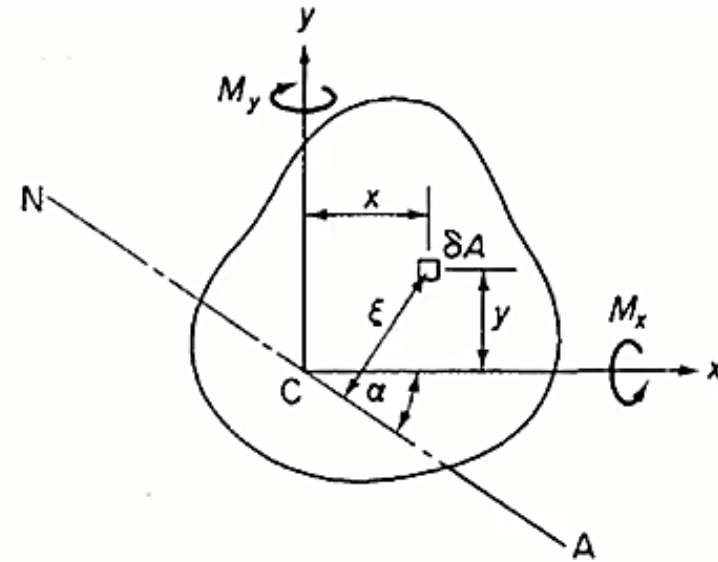
$$M_x = \int_A \sigma_z y \, dA,$$

$$\sigma_z = E \frac{(x \sin \alpha + y \cos \alpha)}{\rho}$$

$$M_x = \int E \frac{(x \sin \alpha + y \cos \alpha)}{\rho} y \, dA$$

$$= \int E \frac{(xy \sin \alpha + y^2 \cos \alpha)}{\rho} \, dA$$

$$M_x = E \frac{(I_{xy} \sin \alpha + I_{xx} \cos \alpha)}{\rho}$$



$$I_{xx} = \int_A y^2 \, dA$$

$$I_{yy} = \int_A x^2 \, dA$$

$$I_{xy} = \int_A xy \, dA$$

$$M_y = \int_A \sigma_z x \, dA$$

$$M_y = \int E \frac{(x \sin \alpha + y \cos \alpha)}{\rho} x \, dA$$

$$= \int E \frac{(x^2 \sin \alpha + xy \cos \alpha)}{\rho} \, dA$$

$$M_y = E \frac{(I_{yy} \sin \alpha + I_{xx} \cos \alpha)}{\rho}$$

$$M_x = \frac{E \sin \alpha}{\rho} I_{xy} + \frac{E \cos \alpha}{\rho} I_{xx}$$

$$M_y = \frac{E \sin \alpha}{\rho} I_{yy} + \frac{E \cos \alpha}{\rho} I_{xy}$$

$$\begin{Bmatrix} M_x \\ M_y \end{Bmatrix} = \frac{E}{\rho} \begin{bmatrix} I_{xy} & I_{xx} \\ I_{yy} & I_{xy} \end{bmatrix} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix}$$

$$\frac{E}{\rho} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix} = \begin{bmatrix} I_{xy} & I_{xx} \\ I_{yy} & I_{xy} \end{bmatrix}^{-1} \begin{Bmatrix} M_x \\ M_y \end{Bmatrix}$$

$$\frac{E}{\rho} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix} = \frac{1}{I_{xx}I_{yy} - I_{xy}^2} \begin{bmatrix} -I_{xy} & I_{xx} \\ I_{yy} & -I_{xy} \end{bmatrix} \begin{Bmatrix} M_x \\ M_y \end{Bmatrix}$$

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

If $M_y = 0$, M_x will produce a stress which varies with respect to x and y .

Similarly, it happens, if $M_x = 0$

In the case, where the beam cross-section has either (or both) X or Y as an axis of symmetry, the product second moment of area I_{xy} is zero.



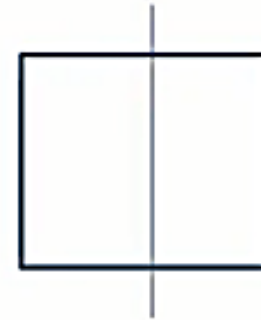
$$\sigma_z = 0$$

$$y = 0$$

x axis becomes the neutral axis when $M_y = 0$

$$\sigma_z = \frac{M_x}{I_{xx}} y + \frac{M_y}{I_{yy}} x$$

$$\sigma_z = \frac{M_x}{I_{xx}} y \quad \text{or} \quad \sigma_z = \frac{M_y}{I_{yy}} x$$



$$\sigma_z = 0$$

$$x = 0$$

y axis becomes the neutral axis when $M_x = 0$

position of the neutral axis depends on the form of the applied loading as well as the geometrical properties of the cross-section.

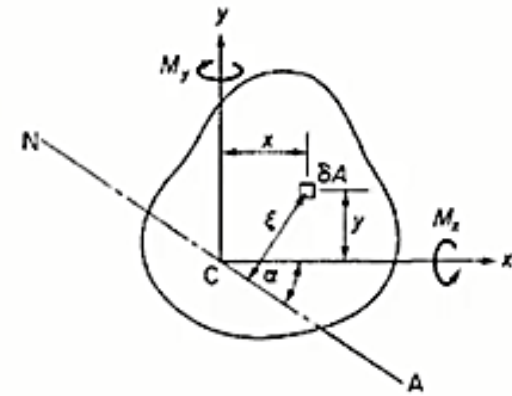
Position of Neutral Axis

The neutral axis always passes through the centroid of area of a beam's cross-section but its inclination α to the x axis depends on the form of the applied loading and the geometrical properties of the beam's cross-section.

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$0 = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x_{NA} + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y_{NA}$$

where x_{NA} and y_{NA} are the coordinates of any point on the neutral axis.



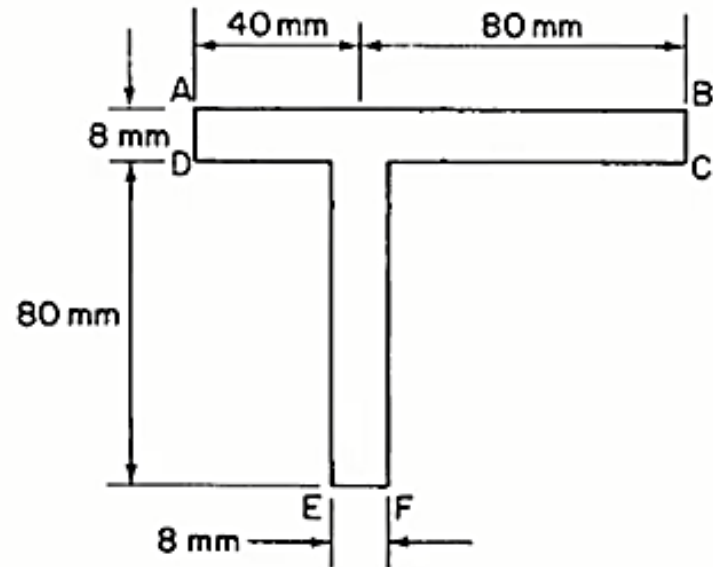
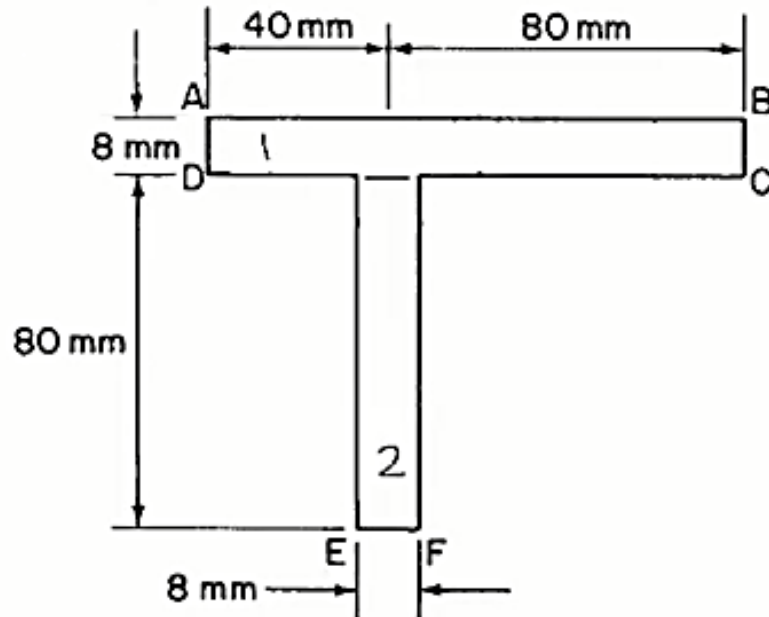
$$\frac{y_{NA}}{x_{NA}} = -\frac{M_y I_{xx} - M_x I_{xy}}{M_x I_{yy} - M_y I_{xy}} \quad \tan \alpha = \frac{M_y I_{xx} - M_x I_{xy}}{M_x I_{yy} - M_y I_{xy}}$$

For a symmetric section where I_{xy} is zero,

$$\tan \alpha = \frac{M_y I_{xx}}{M_x I_{yy}}$$

Example

A beam having the cross-section shown in Fig. is subjected to a bending moment of 1500 N m in a vertical plane. Calculate the maximum direct stress due to bending stating the point at which it acts.



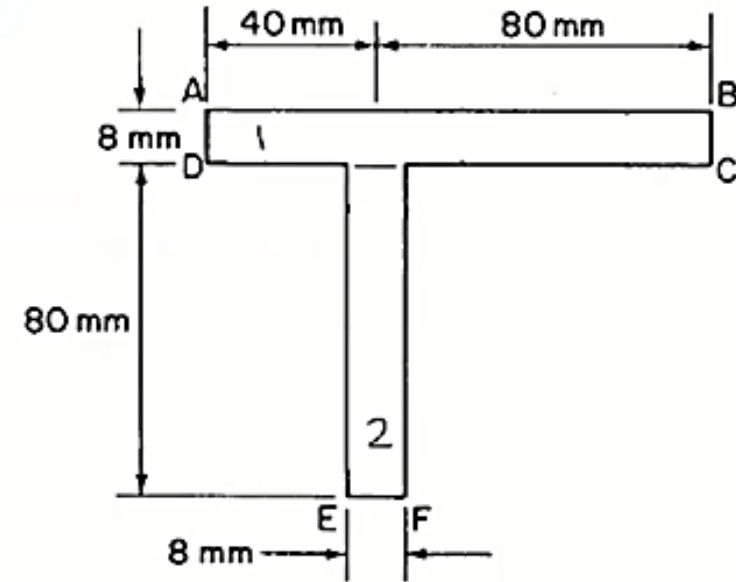
$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

In the problem, the BM in a vertical plane is given, which means M_x is given and M_y is zero. Therefore, the above expression reduces to

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$\sigma_z = \left(\frac{-M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$\sigma_z = \left(\frac{-M_x I_{xy} x + M_x I_{yy} y}{I_{xx} I_{yy} - I_{xy}^2} \right)$$



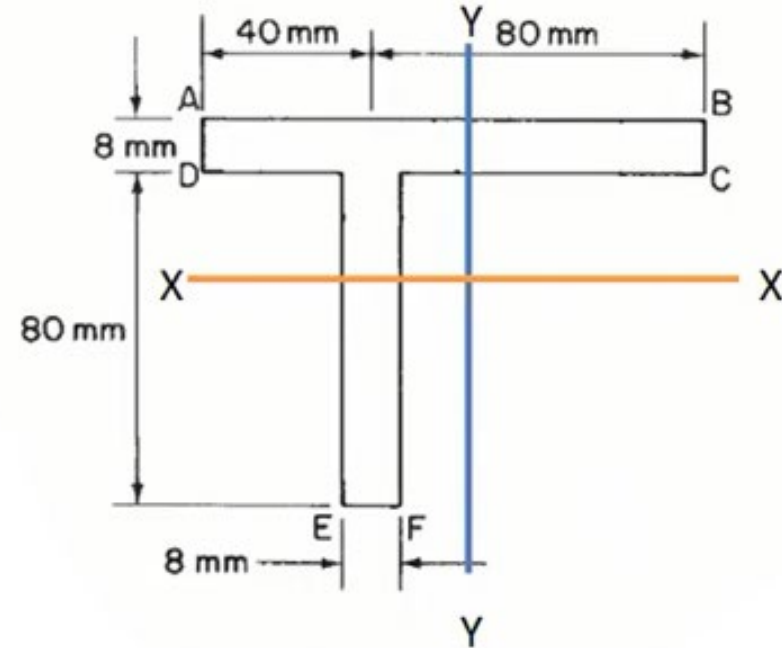
Divide the given section into two rectangles 1 and 2. Take E as reference point and find x (horizontal centroid distance of the individual section from E) and y (vertical centroid distance of the individual section from E)

$$A_1 = 120 \times 8 = 960 \text{ mm}^2: \quad x_1 = 24 \text{ mm}; \quad x_2 = 4 \text{ mm}$$

$$A_2 = 8 \times 80 = 640 \text{ mm}^2: \quad y_1 = 84 \text{ mm}; \quad y_2 = 40 \text{ mm}$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = 16 \text{ mm (location of Y-Y axis from E)}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = 66.4 \text{ mm (location of X-X axis from E)}$$



$$I_{xx} = \frac{bd^3}{12} + A(y - \bar{y})^2 = 302489.6 + 787387.733 = 1.089 \times 10^6 \text{ mm}^4$$

$$I_{yy} = \frac{db^3}{12} + A(x - \bar{x})^2 = 1213440 + 95573.333 = 1.309 \times 10^6 \text{ mm}^4$$

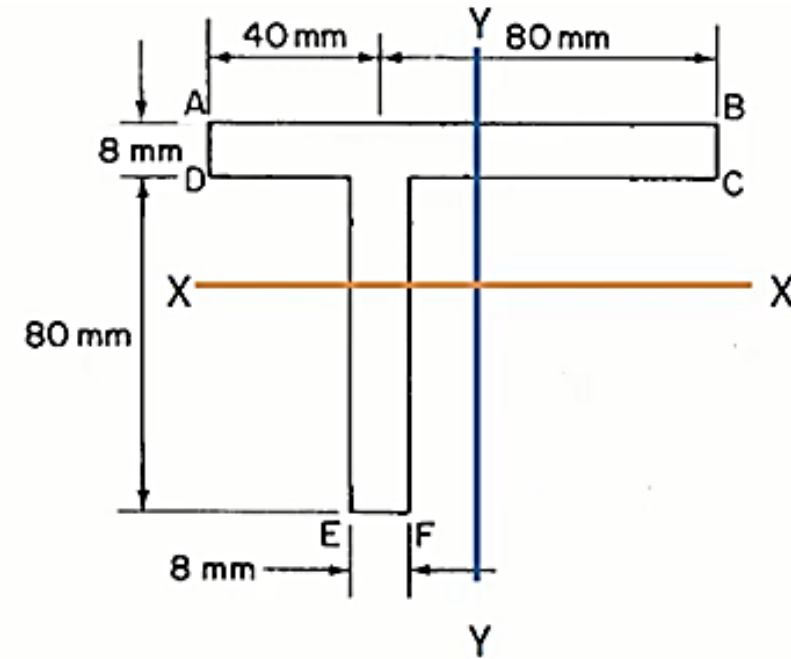
$$I_{xy} = A(x - \bar{x})(y - \bar{y}) = 135168 + 202752 = 0.337 \times 10^6 \text{ mm}^4$$

$$M_x = 1500 \text{ Nm} = 1.5 \times 10^6 \text{ Nmm}$$

$$\sigma_z = \left(\frac{-M_x I_{xy} x + M_x I_{yy} y}{I_{xx} I_{yy} - I_{xy}^2} \right)$$

$$\sigma_z = -0.385 x + 1.495 y$$

$$\sigma_z = -0.385 x + 1.495 y$$



Points	X (mm)	Y(mm)	σ_z (N/mm ²)
A	-52	21.6	52.312
B	68	21.6	6.112
C	68	13.6	-5.848
D	-52	13.6	40.352
E	-16	-66.4	-93.108
F	-8	-66.4	-96.188

Example

Figure shows the section of an angle purlin. A bending moment of 3000 N m is applied to the purlin in a plane at an angle of 30° to the vertical y axis. If the sense of the bending moment is such that its components M_x and M_y both produce tension in the positive xy quadrant, calculate the maximum direct stress in the purlin stating clearly the point at which it acts.

SOLUTION:

Divide the given section into two rectangles 1 and 2. Take C as reference point and find x (horizontal centroid distance of the individual section from C) and y (vertical centroid distance of the individual section from C)

$$A_1 = 100 \times 10 = 1000 \text{ mm}^2:$$

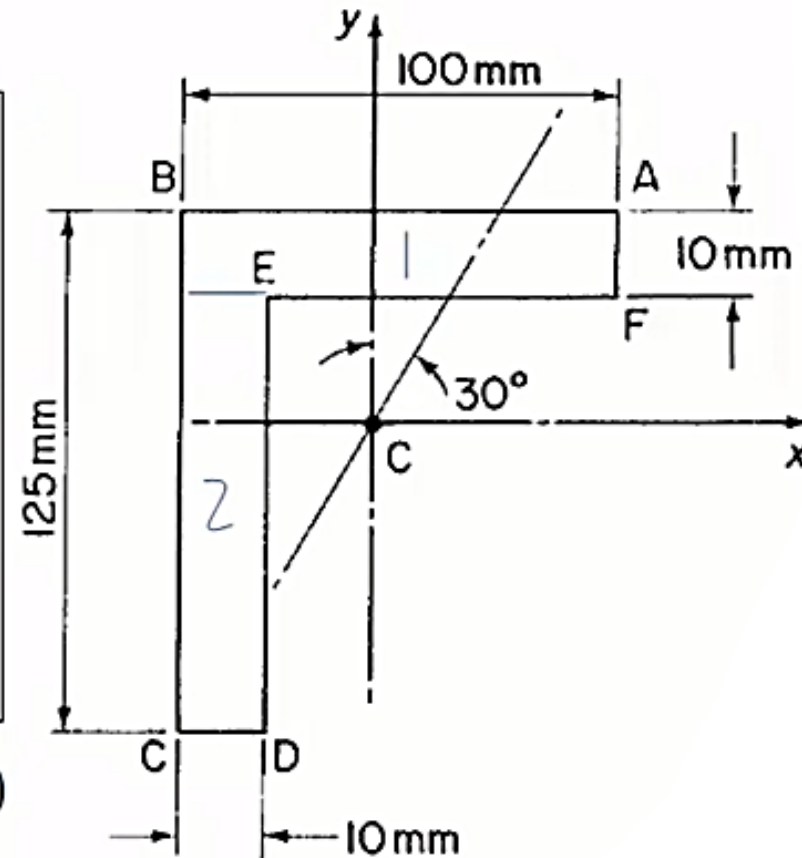
$$x_1 = 50 \text{ mm}; \quad x_2 = 5 \text{ mm}$$

$$A_2 = 115 \times 10 = 1150 \text{ mm}^2:$$

$$y_1 = 120 \text{ mm}; \quad y_2 = 57.5 \text{ mm}$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = 25.93 \text{ mm (location of Y-Y axis from C)}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = 86.57 \text{ mm (location of X-X axis from C)}$$



$$I_{xx} = \frac{bd^3}{12} + A(y - \bar{y})^2 = 1125898 + 2239220 = 3.365 \times 10^6 \text{ mm}^4$$

$$I_{yy} = \frac{db^3}{12} + A(x - \bar{x})^2 = 1412698 + 513357.968 = 1.926 \times 10^6 \text{ mm}^4$$

$$I_{xy} = A(x - \bar{x})(y - \bar{y}) = 804660 + 699700.365 = 1.504 \times 10^6 \text{ mm}^4$$

$$M = 3000 \text{ Nm} = 3 \times 10^6 \text{ Nmm}$$

$$M_x = M \cos(30) = 2.598 \times 10^6 \text{ Nmm}$$

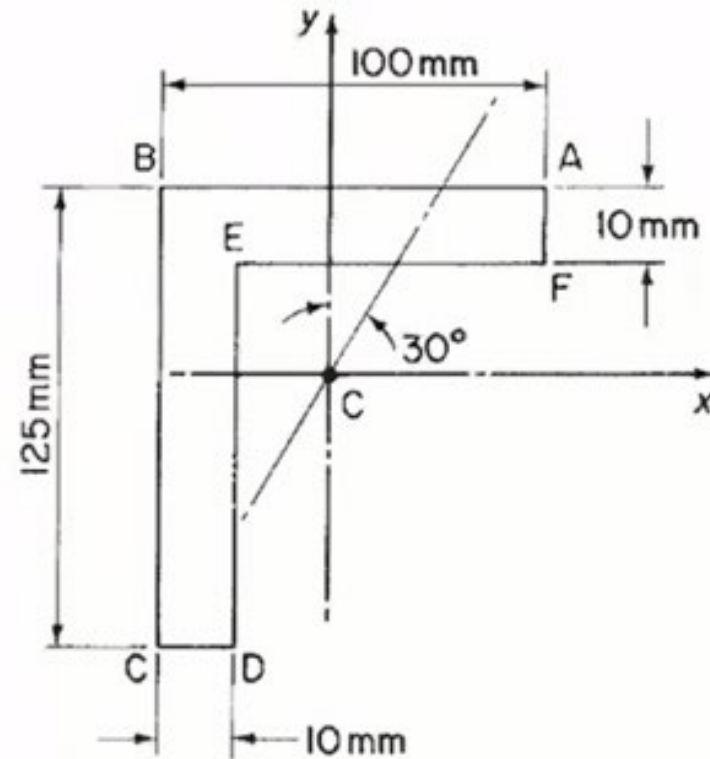
$$M_y = M \sin(30) = 1.5 \times 10^6 \text{ Nmm}$$

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$\sigma_z = 0.27x + 0.65y$$

At C, $x = -25.93 \text{ mm}$, $y = -86.57 \text{ mm}$,

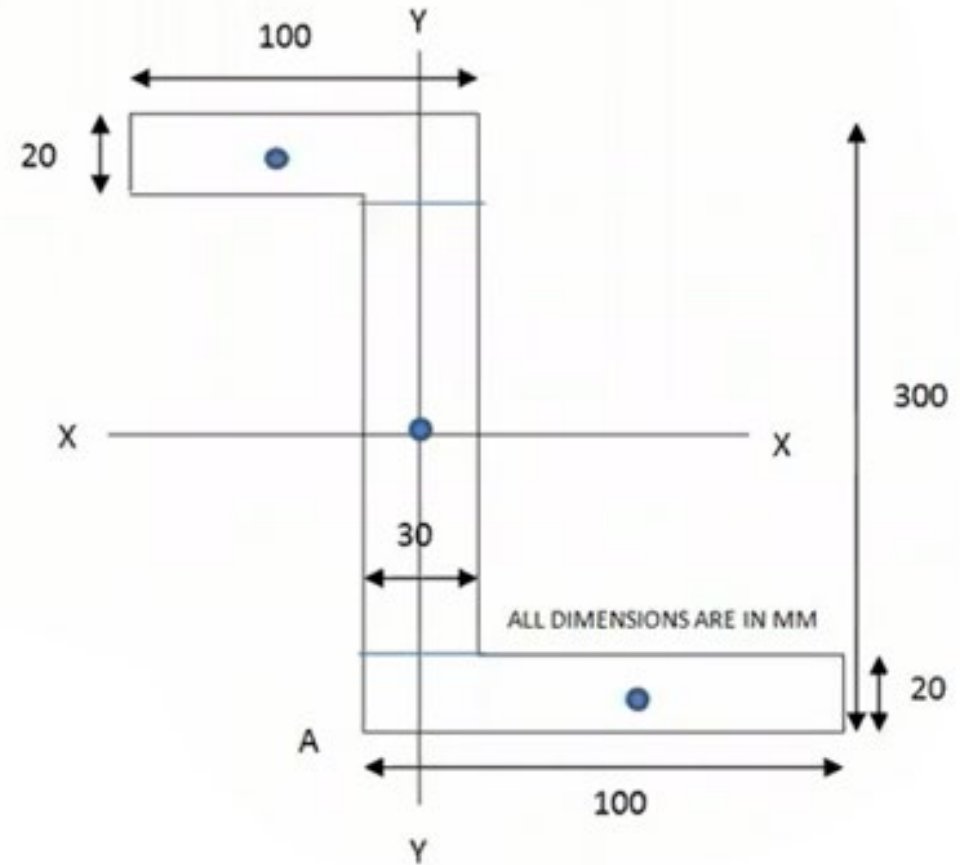
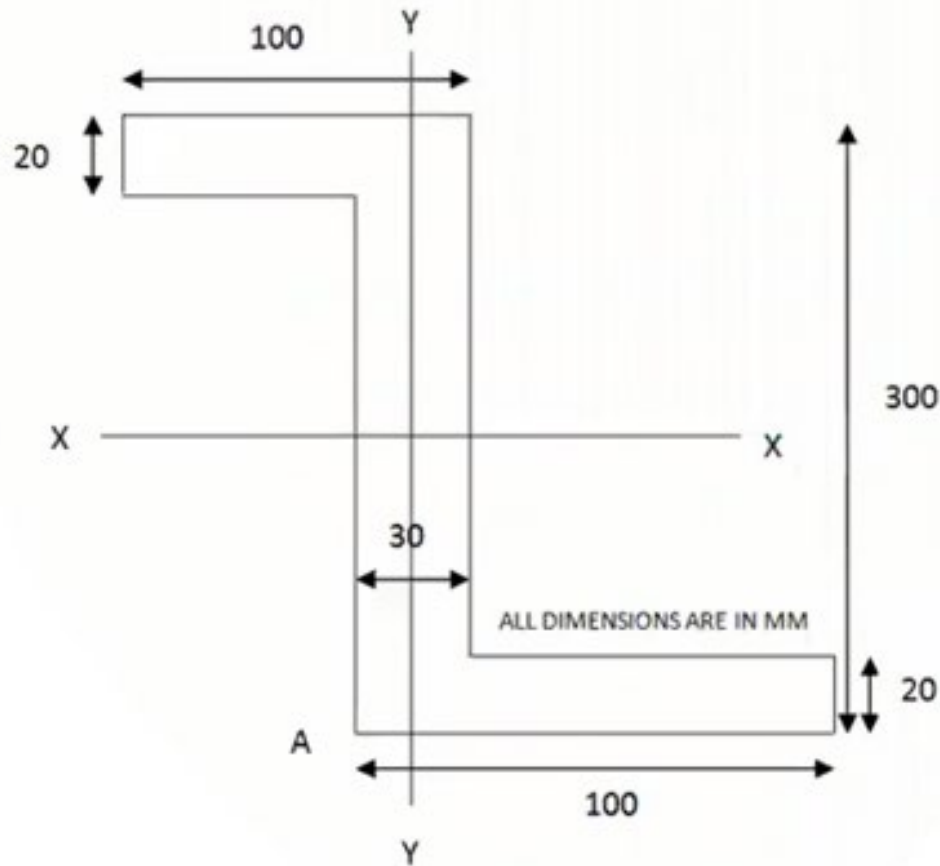
$$\sigma_z = -63.27 \text{ N/mm}^2$$



Example

Find the direct stress at A due to bending when $M_x = 100\text{kNm}$, $M_y = 50\text{ kNm}$.

SOLUTION:



$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$I_{xx} = \frac{bd^3}{12} + A(y - \bar{y})^2 = 39266667 + 39266667 + 43940000 = 122.473 \times 10^6 \text{ mm}^4$$

$$I_{yy} = \frac{db^3}{12} + A(x - \bar{x})^2 = 4116667 + 4116667 + 585000 = 8.818 \times 10^6 \text{ mm}^4$$

$$I_{xy} = A(x - \bar{x})(y - \bar{y}) = -9800000 - 9800000 = -19.6 \times 10^6 \text{ mm}^4$$

$$M_x = 100 \text{ kNm} = 100 \times 10^6 \text{ Nmm}$$

$$M_y = 50 \text{ kNm} = 50 \times 10^6 \text{ Nmm}$$

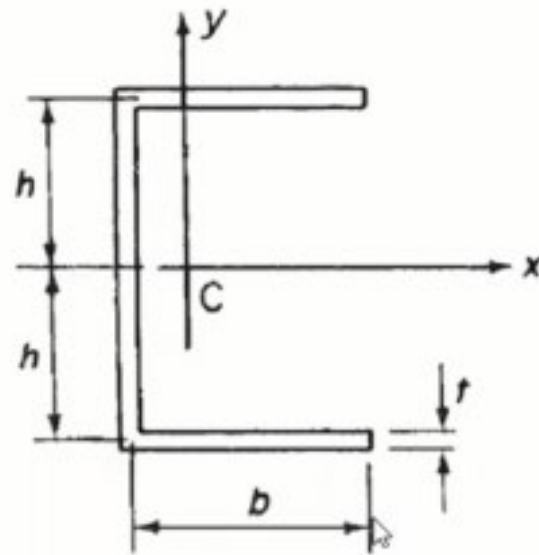
$$\sigma_z = 1.16E - 05x + 2.67E - 06y$$

At A, $x = -15 \text{ mm}$, $y = -150 \text{ mm}$,

$$\sigma_z = -0.00058 \text{ N/mm}^2$$

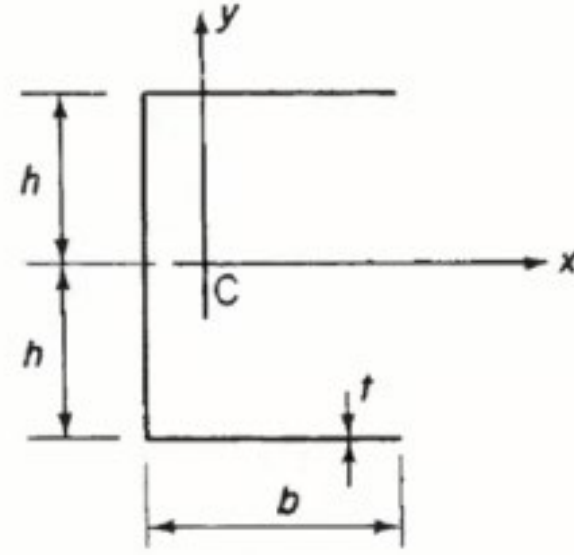
Approximations of thin walled sections:

- Some engineering structures are thin-walled in nature, hence assumptions can be simplified in the determination of stresses and deflections produced by bending.
- The thickness “ t ” of thin-walled sections is assumed to be small compared with their cross sectional dimensions.
- The stresses may be considered as a constant across the thickness.
- Therefore, we can neglect squares and higher powers of “ t ” in the computation of sectional properties.
- The resulting section is represented by the mid-line of its wall.



(a)

(a) Actual thin-walled channel section



(b)

(b) approximate representation of section.

$$I_{xx} = 2 \left[\frac{bt^3}{12} + (bt)(h^2) \right] + \frac{t(2h)^3}{12}$$

$$I_{xx} = 2 \left[\frac{bt^3}{12} + (bt)(h^2) \right] + \frac{t(2h)^3}{12}$$

$$I_{xx} = 2(bt)(h^2) + \frac{2t(h)^3}{3}$$

Actual:

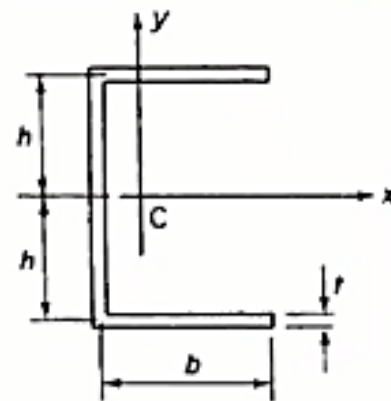
$$t = 2 \text{ mm}; b = 10 \text{ mm}; h = 12 \text{ mm}$$

$$I_{xx} = 2 \left[\frac{(10)(2^3)}{12} + (10 \times 2)(12^2) \right] + \frac{2(2 \times 12)^3}{12} = 8077.333 \text{ mm}^4$$

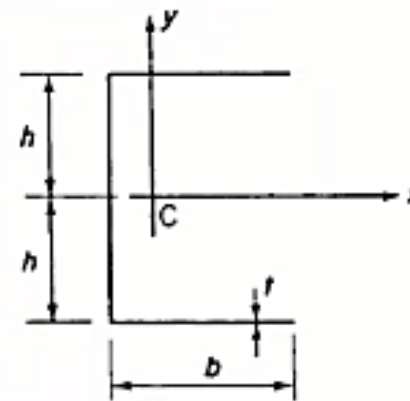
Approximate:

$$t = 2 \text{ mm}; b = 10 \text{ mm}; h = 12 \text{ mm}$$

$$I_{xx} = 2(10 \times 2)(12^2) + \frac{2 \times 2(12)^3}{3} = 8064 \text{ mm}^4$$



(a)
(a) Actual thin-walled channel section



(b)
(b) approximate representation of section.



Thank you

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